Software Design Document (SDD)
Enhancement to PostgreSQL COPY Command

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Prepared by: Group 7
Project: Enhancing PostgreSQL COPY Function

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1. Introduction
1.1 Purpose
This document outlines the design for enhancing PostgreSQL's COPY function with additional capabilities such as data preview, pre-validation, parallel copy operations, and extended support for multiple file formats. These improvements aim to optimize data import/export, enhance usability, and improve compatibility with modern data workflows.

1.2 Scope
The system:

- Implements multi-threaded execution for parallel copy operations.
- Provides data preview capabilities before execution.
- Supports additional file formats such as JSON, JSONL and Avro.
- Ensures pre-validation checks for data integrity before insertion.

1.3 Definitions & Acronyms
- **COPY:** A PostgreSQL command for importing/exporting large datasets.
- **PostgreSQL:** An open-source relational database management system.
- **JSON/JSONL:** Lightweight formats for structured data representation.
- **Avro:** Columnar storage format for big data applications.

1.4 References

1. PostgreSQL Documentation
2. GitHub PostgreSQL Repository

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2. Problem Statement & Proposed Enhancements

2.1 Problem Statement

The existing **PostgreSQL COPY** command has the following limitations:

- **Lack of Data Preview:** Users cannot view data before importing.
- **Manual Pre-Validation:** Schema and format validation must be handled separately.
- **Single-Threaded Execution:** COPY operations are **not parallelized**, limiting performance.
- **No Batch Processing:** Large data files must be imported as a single operation.
- **Limited File Format Support:** COPY does not support JSON, JSONL and Avro.

2.2 Proposed Enhancements

- **Parallel Processing** – Multi-threaded COPY execution to speed up imports. **Batch**
- **Processing** – Splitting large files into smaller chunks for optimized loading. **Data**
- **Preview** – Allow users to preview a few rows before execution. **Pre-Validation**
- **Checks** – Schema and format validation before copying. **Extended File Format**
- **Support** – Native support for **JSON, Avro, JSONL**.

  Improved Error Handling – Automatic rollback and logging of failures.

3. System Overview

3.1 High-Level Architecture

The system comprises:

- **Data Processing:** Prepares data for the COPY operation, including format validation.
- **Parallel Execution Engine:** Handles multi-threaded processing of data transfer.
- **Pre-validation Module:** Performs integrity checks before execution.
- **File Format Handler:** Extends COPY support to additional file formats.

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4. System Components

4.1 Functional Components

Component	Technology	Description
Data Processor	PostgreSQL,C	Prepares and validates input data for COPY
Parallel Engine	Multi-threading, C	Adds support for additional formats

Format Handler	PostgreSQL Extension,C	Adds support for additional formats
Validation Engine	PosgreSQL	Simple UI to test performance

4.2 Subsystems & Modules

1. **Parallel Execution Module:**
    - **Implements multi-threaded processing for improved performance.**
2. **Batch Processing Module:**
    - Splits large data files into manageable chunks.
3. **Error Handling & Recovery:**
    - Provides rollback mechanisms and retry operations.
4. **Security Features:**
    - Implements role-based access control (RBAC) and encryption for secure data transfers.

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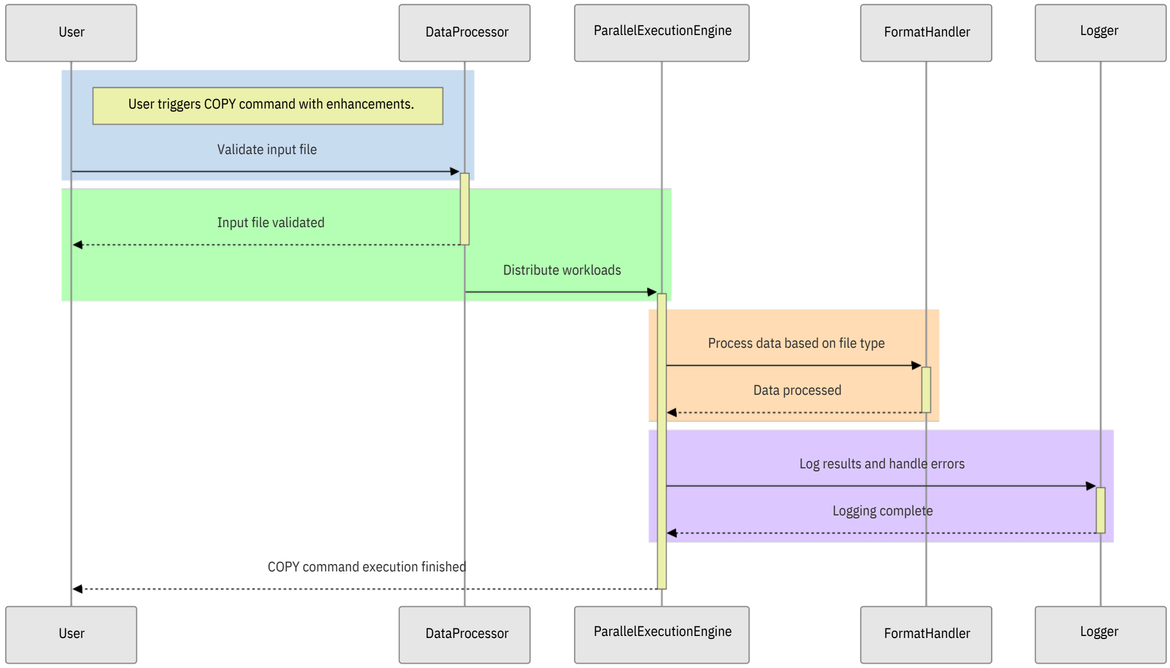
4. Workflow & Interactions

4.1 Sequence Diagram



5.Data Design

5.1 Data design For File Format

```psql
COPY employees
FROM '/home/amalendu/college/employees.json'
WITH (FORMAT 'json');
```

5.2 Data Model for Data Preview Feature

```psql
COPY PREVIEW employees
FROM '/home/amalendu/college/employees.json'
WITH (FORMAT 'json', ROWS 5);
```

6. Security Considerations

6.1 Data Privacy

- Encrypts COPY transactions.
- Implements RBAC for access control.

6.2 Model Security

- Logs all operations for auditability.
- Ensures rollback on failed transactions.

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7. Testing Strategy

7.1 Unit Testing

Test Type	Tools Used
COPY Performance	PostgreSQL Benchmarks

7.2 Integration Testing

Component	Tool Used
Parallel Execution	Docker

8. Deployment Strategy

Component	Deployment Environment
COPY Enhancements	PostgreSQL Plugin
Validation Engine	PostgreSQL Stored Procedures

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9. Compliance & Standards

- Ensures compliance with ACID properties of PostgreSQL.
- Adheres to best practices for database performance optimization.

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10. Conclusion

This SDD describes a scalable, efficient enhancement to PostgreSQL's COPY function, improving performance, security, and data format compatibility. These features will enhance PostgreSQL’s suitability for large-scale data workloads and analytics.
